

Python Code

```
1
2  **Project Name:** Number Guessing Game
3  **Developer:** [Your Name Here]
4  **Language Used:** Python 3.x
5
6  ---
7
8  ## 🎯 1. Objective
9  The primary objective of this project is to practically implement core Python
10 programming concepts, user input validation, and exception handling. This
11 project demonstrates how to build a robust, error-free console game.
12
13 ## 💻 2. Core Logic & Code Structure
14 The application utilizes the following essential programming concepts:
15 - **`random.randint(1, 100)`**: Automatically generates a secure, random
16   secret number between 1 and 100 at the start of the game.
17 - **`while True` Loop**: Keeps the game running continuously until the user
18   successfully guesses the correct number.
19 - **`attempts += 1`**: Tracks and increments the user's total guesses to
20   display their final score upon winning.
21
22 ## 🛡️ 3. Industry Standard Best Practices
23 To ensure the software is reliable and ready for professional review, the code
24 includes:
25 - **Robust Error Handling**: Implements a `try-except` block to catch
26   `ValueError`. If a user accidentally inputs text (e.g., "abc") or symbols
27   instead of a number, the program handles it gracefully without crashing and
28   prompts the user to enter a valid whole number.
29
30 ## 🚀 4. Evaluation for Recruiters
31 This project demonstrates a strong foundational understanding of:
32 - Python syntax and logical flow control (`if-elif-else`).
33 - Working with built-in modules (`random`).
34 - Writing clean, maintainable, and user-centric code with proper exception
35   handling.
```